

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 3 – MCQ Test

|                  |                                                                                                                                |               |                              |
|------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------|
| Course Code:     | DSA03                                                                                                                          | Course Title: | Natural Language Processing  |
| CO Assessed:     | CO3 – Precision                                                                                                                | Assessment:   | Assessment Tool 3 – MCQ Test |
| Weightage:       | 10% of CIA                                                                                                                     |               |                              |
| CO5 Statement:   | Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3) |               |                              |
| Student Name:    | Y Greeshma Reddy                                                                                                               | Reg. No.:     | 192424146                    |
| Section / Batch: | B Tech-AI&DS                                                                                                                   | Date:         | 09-08-2026                   |

### Code of Conduct

*I Y Greeshma Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Y Greeshma Reddy

### Instructions

- Read each question carefully before selecting your answer.
- This assessment consists of 50 multiple-choice questions.
- Do not use external resources, AI tools, textbooks, or assistance from others during the assessment unless explicitly permitted by the instructor.
- Upload Score Card in GCR and submit score card printout.
- Duration: 30 minutes.

| Section / Topic                            | Focus                                                                                 | Q Nos   |
|--------------------------------------------|---------------------------------------------------------------------------------------|---------|
| Context-Free Grammars (CFG)                | CFG components, productions, terminals, non-terminals, recursion, sentence generation | 1 – 10  |
| Sentence Construction and Agreement        | Parse trees, sentence structure, grammatical agreement, syntactic validation          | 11 – 20 |
| Subcategorization                          | Verb frames, complement requirements, argument structure analysis                     | 21 – 25 |
| Top-Down Parsing                           | Parsing strategy, derivation process, recursion issues, grammar expansion             | 26 – 30 |
| Earley Parsing                             | Predictor, Scanner, Completer operations, ambiguity handling                          | 31 – 35 |
| Feature Structures                         | Feature representation, unification, constraint checking, agreement enforcement       | 36 – 40 |
| Probabilistic Context-Free Grammars (PCFG) | Rule probabilities, ambiguity resolution, probabilistic parsing                       | 41 – 48 |
| Integrated Syntax Analysis                 | CFG + Feature Structures + PCFG integration in NLP applications                       | 49 – 50 |

<https://forms.gle/AjecXtqbtwvUYbYA9>



**Attach score card screenshot:**

## Rubrics for Calculation

| Criteria                  | Weight age | Level 4 – Exemplary      | Level 3 – Proficient     | Level 2 – Developing     | Level 1 – Beginning     |
|---------------------------|------------|--------------------------|--------------------------|--------------------------|-------------------------|
| Number of Correct Answers | 100%       | 45 to 50 correct answers | 38 to 44 correct answers | 25 to 37 correct answers | 0 to 24 correct answers |
| Accuracy                  | 100%       | Excellent (90–100%)      | Good (75–89%)            | Average (50–74%)         | Poor (Below 50%)        |

| No. of Questions | Number of Correct Answers | Accuracy | Total % |
|------------------|---------------------------|----------|---------|
| 50               |                           |          |         |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
|                   |                    |                  |
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |

## UNIT I – Introduction and Morphology

**1. What is the primary goal of Natural Language Processing (NLP)?**

- A) To increase computer hardware speed
- B) To enable computers to process and understand human language
- C) To design operating systems
- D) To manage computer networks

**Answer: B) To enable computers to process and understand human language**

**2. Which model is commonly used to describe whether a string belongs to a particular language?**

- A) Database model
- B) Mathematical model
- C) Financial model
- D) Hardware model

**Answer: B) Mathematical model**

**3. Which symbol is commonly used in regular expressions to represent zero or more occurrences of the preceding element?**

- A) +
- B) ?

C) \*

D) ^

**Answer: C) \***

**4. Which regular expression represents a sequence containing one or more occurrences of a?**

A) a\*

B) a+

C) a?

D) a|

**Answer: B) a+**

**5. What is the main function of a Finite State Automaton (FSA)?**

A) To store large databases

B) To recognize patterns in strings

C) To translate programming languages

D) To generate images

**Answer: B) To recognize patterns in strings**

**6. Which type of morphology deals with grammatical variations such as tense and number?**

A) Derivational morphology

B) Inflectional morphology

C) Computational morphology

D) Structural morphology

**Answer: B) Inflectional morphology**

**7. Which example represents derivational morphology?**

A) cat → cats

B) walk → walked

C) happy → happiness

D) book → books

**Answer: C) happy → happiness**

**8. What is the purpose of finite-state morphological parsing?**

A) To identify the internal morphological structure of words

B) To identify network protocols

C) To calculate sentence length

D) To generate numerical data

**Answer: A) To identify the internal morphological structure of words**

**9. What does the Porter Stemmer primarily perform?**

A) Sentence parsing

B) Word stemming

- C) POS tagging
- D) Speech recognition

**Answer: B) Word stemming**

**10. The word “unhappiness” can be analyzed as un + happy + ness. Which process is illustrated by this analysis?**

- A) Inflectional morphology only
- B) Derivational morphology
- C) Syntax parsing
- D) Regular expression matching

**Answer: B) Derivational morphology**

## **UNIT II – Word Level and Syntactic Analysis**

**11. What does an N-gram model estimate?**

- A) Probability of a word based on previous words
- B) Number of letters in a word
- C) Meaning of a paragraph
- D) Number of sentences in a document

**Answer: A) Probability of a word based on previous words**

**12. A bigram model considers how many words at a time?**

- A) One
- B) Two
- C) Three
- D) Four

**Answer: B) Two**

**13. What is the main problem with an unsmoothed N-gram model?**

- A) It cannot count words
- B) It assigns zero probability to unseen N-grams
- C) It cannot recognize nouns
- D) It cannot process sentences

**Answer: B) It assigns zero probability to unseen N-grams**

**14. What is the primary purpose of smoothing in language models?**

- A) To remove punctuation
- B) To assign probabilities to unseen events
- C) To identify verbs
- D) To shorten sentences

**Answer: B) To assign probabilities to unseen events**

**15. What does entropy measure in an N-gram language model?**

- A) Linguistic uncertainty
- B) Number of nouns
- C) Sentence length
- D) Number of characters

**Answer: A) Linguistic uncertainty**

**16. Which word class does “quickly” belong to in the sentence “She runs quickly”?**

- A) Noun
- B) Verb
- C) Adjective
- D) Adverb

**Answer: D) Adverb**

**17. What is the purpose of a tagset in POS tagging?**

- A) To define the set of grammatical labels used for words
- B) To store sentences
- C) To generate regular expressions
- D) To calculate entropy

**Answer: A) To define the set of grammatical labels used for words**

**18. In the sentence “Book a ticket,” what is the POS of “Book”?**

- A) Noun
- B) Verb
- C) Adjective
- D) Adverb

**Answer: B) Verb**

**19. Which POS tagging approach uses manually written linguistic rules?**

- A) Stochastic tagging
- B) Rule-based tagging
- C) Transformation-based tagging
- D) Random tagging

**Answer: B) Rule-based tagging**

**20. Which POS tagging method combines an initial tagging with a sequence of corrective transformations?**

- A) Rule-based tagging
- B) Stochastic tagging
- C) Transformation-based tagging

D) Dictionary-based tagging

**Answer: C) Transformation-based tagging**

## **UNIT III – Context-Free Grammars**

**21. What is the primary purpose of a Context-Free Grammar (CFG) in NLP?**

A) To represent syntactic structure of sentences

B) To calculate word frequency

C) To identify word meanings

D) To translate documents

**Answer: A) To represent syntactic structure of sentences**

**22. Which component of a CFG specifies how a non-terminal can be expanded?**

A) Terminal symbol

B) Production rule

C) Probability table

D) Lexicon only

**Answer: B) Production rule**

**23. In the rule  $S \rightarrow NP VP$ , what does S usually represent?**

A) Sentence

B) Subject

C) Semantics

D) String

**Answer: A) Sentence**

**24. Which structure visually represents the syntactic organization of a sentence?**

A) Bar chart

B) Parse tree

C) Flowchart

D) Frequency table

**Answer: B) Parse tree**

**25. What does grammatical agreement ensure?**

A) Words have the same number of letters

B) Related words have compatible grammatical features

C) All words have the same meaning

D) Sentences contain equal numbers of words

**Answer: B) Related words have compatible grammatical features**

**26. What does subcategorization describe?**

- A) The arguments or complements that a word, especially a verb, requires
- B) The number of letters in a word
- C) The probability of a sentence
- D) The meaning of punctuation

**Answer: A) The arguments or complements that a word, especially a verb, requires**

**27. What is parsing in NLP?**

- A) Converting text into numbers only
- B) Determining the syntactic structure of a sentence
- C) Removing stop words
- D) Translating words between languages

**Answer: B) Determining the syntactic structure of a sentence**

**28. Which parsing strategy begins with the start symbol and attempts to generate the input sentence?**

- A) Bottom-up parsing
- B) Top-down parsing
- C) Stochastic tagging
- D) Semantic parsing

**Answer: B) Top-down parsing**

**29. Which parsing algorithm can parse sentences using dynamic programming and handle left-recursive grammars?**

- A) Porter algorithm
- B) Earley parser
- C) Viterbi tagger
- D) PageRank algorithm

**Answer: B) Earley parser**

**30. What is the purpose of a Probabilistic Context-Free Grammar (PCFG)?**

- A) To associate probabilities with grammar productions
- B) To remove grammar rules
- C) To identify word stems
- D) To classify documents

**Answer: A) To associate probabilities with grammar productions**

## **UNIT IV – Semantic Analysis**

**31. What is the main objective of semantic analysis?**

- A) To determine the meaning of linguistic expressions



- B) To count characters
- C) To identify punctuation
- D) To format documents

**Answer: A) To determine the meaning of linguistic expressions**

**32. Which formal system is commonly used to represent relationships between entities and properties?**

- A) First-Order Predicate Calculus
- B) Regular Expression
- C) Finite Automaton
- D) N-gram model

**Answer: A) First-Order Predicate Calculus**

**33. In first-order predicate logic, what does  $P(x)$  typically represent?**

- A) A probability
- B) A predicate applied to an argument
- C) A parsing rule
- D) A POS tag

**Answer: B) A predicate applied to an argument**

**34. Which representation can express the statement “John loves Mary” as a relationship between entities?**

- A) loves(John, Mary)
- B) John + Mary
- C) John/Mary
- D) John = Mary

**Answer: A) loves(John, Mary)**

**35. What is syntax-driven semantic analysis?**

- A) Deriving meaning representations based on syntactic structure
- B) Counting syntactic words
- C) Removing grammatical rules
- D) Generating random meanings

**Answer: A) Deriving meaning representations based on syntactic structure**

**36. What is a lexeme?**

- A) An abstract vocabulary unit representing related word forms
- B) A punctuation mark
- C) A sentence structure
- D) A grammar rule

**Answer: A) An abstract vocabulary unit representing related word forms**

**37. What does word sense refer to?**

- A) The grammatical category of a word
- B) A particular meaning of a word
- C) The number of letters in a word
- D) The pronunciation of a word

**Answer: B) A particular meaning of a word**

**38. What is the main goal of Word Sense Disambiguation (WSD)?**

- A) To determine the intended meaning of an ambiguous word
- B) To identify sentence boundaries
- C) To count words
- D) To remove prefixes

**Answer: A) To determine the intended meaning of an ambiguous word**

**39. In “The fisherman sat on the bank,” which meaning of “bank” is most appropriate?**

- A) Financial institution
- B) River side
- C) Storage facility
- D) Computer memory

**Answer: B) River side**

**40. What is the primary goal of Information Retrieval (IR)?**

- A) To retrieve relevant information from a collection of documents
- B) To generate grammar rules
- C) To identify speech sounds
- D) To stem every word

**Answer: A) To retrieve relevant information from a collection of documents**

## **UNIT V – Language Generation and Discourse Analysis**

**41. What does discourse analysis primarily study?**

- A) Language beyond individual sentences and its context
- B) Individual characters only
- C) Computer hardware
- D) Word spelling only

**Answer: A) Language beyond individual sentences and its context**

**42. What is reference resolution?**

- A) Determining what an expression such as “he” or “it” refers to
- B) Finding the number of words in a sentence

C) Identifying punctuation marks

D) Translating a sentence

**Answer: A) Determining what an expression such as “he” or “it” refers to**

**43. In the sentence “Ravi bought a car. He likes it,” what does “He” refer to?**

A) Car

B) Ravi

C) Likes

D) It

**Answer: B) Ravi**

**44. What does text coherence refer to?**

A) The meaningful and logical relationship among parts of a text

B) The number of words in a paragraph

C) The spelling of words

D) The number of punctuation marks

**Answer: A) The meaningful and logical relationship among parts of a text**

**45. What is a discourse act or dialogue act?**

A) The communicative function performed by an utterance

B) The number of words in dialogue

C) A grammatical error

D) A punctuation rule

**Answer: A) The communicative function performed by an utterance**

**46. Which of the following is an example of a dialogue act?**

A) Request

B) Noun

C) Prefix

D) Stem

**Answer: A) Request**

**47. What is the main purpose of a conversational agent?**

A) To interact with users through natural language

B) To perform only mathematical calculations

C) To store images

D) To compile source code

**Answer: A) To interact with users through natural language**

**48. What is the purpose of surface realization in language generation?**

A) To convert an abstract representation into natural-language text

B) To identify word senses

- C) To parse a sentence
- D) To calculate N-gram entropy

**Answer: A) To convert an abstract representation into natural-language text**

**49. Which machine translation approach uses an intermediate language-independent representation?**

- A) Direct translation
- B) Interlingua
- C) Word-for-word translation
- D) Rule deletion

**Answer: B) Interlingua**

**50. What do statistical approaches to machine translation primarily learn from?**

- A) Parallel and bilingual language data
- B) Only punctuation marks
- C) Computer hardware specifications
- D) Regular expressions alone

**Answer: A) Parallel and bilingual language data**